

ME 628: Impact Mechanics

Course Code	ME 628
Course Name	Impact Mechanics
L-T-P-C	3-0-0-3
Prerequisites	Mechanics of Solids (ME206) & FEM in Engineering (ME513) or Advanced FEM (ME601)
Intended for	UG/M. Tech/MS/PhD
Distribution	Elective
Approval	recommended by the 39th BoA (Mar 2021)

Course Contents

- **Introduction**
Impact Mechanics - Appreciating Impact Problems in Engineering, Historical Background, Low velocity impact, Principle of dynamics, Percussion, Concussion, Collision and Explosion. [4 Lectures]
- **Rigid Body Impact Mechanics**
Impulse - Momentum Equations, Measurement of Coefficient of Restitution, Kinetic Coefficient of Restitution, Energy Coefficient of Restitution, Oblique Impact- Role of Friction. [5 Lectures]
- **Impact Mechanics of Deformable Bodies**
One-Dimensional Impact Mechanics of Deformable Bodies: Single Degree of Freedom Idealization of Impact Process, 1-D Wave Propagation in Solids Induced by Impact, Governing Equation for Flexural Waves in Rods, D'Alembert's Solution for Wave Equation.
Multi-Dimensional Impact Mechanics of Deformable Bodies: Analysis of Stress and Strain, Linearised Stress-Strain Relations. [8 Lectures]
- **Experimental Impact Mechanics**
Quasi-Static Material Tests, Pendulum Impact Tests, High Strain Rate Testing of elastoplastic and visco-elastic materials, Split Hopkinson's Pressure Bar Test, Taylor Cylinder Impact Test, Drop Impact Test. [10 Lectures]
- **Impact Modeling**
Modeling Deformation and Failure Under Impact: Equation of State, Gruneisen Parameter and Murnaghan Equation of State, Constitutive Models for Material Deformation and Plasticity, Johnson-Cook Material Model, Determination of Parameters in Johnson-Cook Model, Implementation of Damage, Implementation of penetration and bulging. [4 Lectures]
- **Computational Impact Mechanics**
Lagrangian Formulation, Eulerian Formulation, Computational Aspects in Numerical Simulation, Shockwaves. [3 Lectures]
- **Case Studies**
Low-velocity impact properties of composite materials, Simulation of Ballistic Impact of a Steel Bullet, Mechanics of Vehicle Collision, Case studies of vehicle crash impact test, Case studies of explosive impact test. [7 Lectures]

Text Books

1. C. Lakshmana Rao, V. Narayanamurthy, K. R. Y. Simha, **Applied Impact Mechanics**, John Wiley & Sons Ltd., 2016.
2. W. J. Stronge, **Impact Mechanics**, Cambridge University Press, 2018.

References

1. Laursen, Tod A, **Computational Contact and Impact Mechanics**, Springer. 2003.
2. Ian G. Crouch, **The Science of Armour Materials**, Elsevier, 2016.
3. Karl F. Graff, **Wave Motion in Elastic Solids**, Dover Publications, 2003.
4. Zvi Rosenberg and Erez Dekel, **Terminal Ballistics**, Spinger, 2012.
5. Jonas A Zukas, **High Velocity Impact Dynamics**, John Wiley & Sons. 1990.